

BIOL 250 · Axial HEAD & NECK — PAPER NOTES

QUICK-SCAN INDEX (alphabetical — scan by eye; all key terms)

Scan this page first under time pressure. Full detail in sections below.

TERM	QUICK FACT
Anterior digastric	CN V3 → depresses mandible, elevates hyoid (post. belly = CN VII)
Buccinator	CN VII → compresses cheeks; NOT a mastication muscle
Cricothyroid m.	CN X (superior laryngeal n.) → only laryngeal m. NOT via RLN; tenses vocal folds
Genioglossus	CN XII → protrudes tongue; largest extrinsic tongue m.
Geniohyoid	C1 via CN XII → elevates hyoid, depresses mandible
Hyoglossus	CN XII → depresses + retracts tongue
Inferior oblique (eye)	CN III → elevates + abducts + extorts eyeball
Lateral pterygoid	CN V3 → OPENS mouth / protrudes jaw (only jaw opener)
Lateral rectus (eye)	CN VI (abducens) → abducts eyeball; damage → medial drift (esotropia)
Levator palpebrae	CN III → raises upper eyelid; damage → ptosis
Masseter	CN V3 → elevates mandible (closes jaw); most powerful jaw closer
Medial pterygoid	CN V3 → elevates + deviates mandible medially
Mylohyoid	CN V3 → elevates floor of mouth; depresses mandible
Occipitofrontalis	CN VII → raises eyebrows, wrinkles forehead
Omohyoid	Ansa cervicalis (C1–C3) → depresses hyoid; 2 bellies
Orbicularis oculi	CN VII → closes eyelid; blinks
Orbicularis oris	CN VII → closes/purses lips
Palatoglossus	CN X (vagus) → EXCEPTION: tongue muscle NOT via CN XII
Pharyngeal constrictors	CN X → constrict pharynx; propel bolus
Platysma	CN VII → depresses mandible; wrinkles neck skin
Post. cricoarytenoid	CN X (RLN) → ONLY abductor of vocal folds
SCM (sternocleidomastoid)	CN XI + C2–C3 → flex + rotate head contralateral
Stapedius	CN VII → dampens ossicle vibration (loudest sound reflex)
Sternohyoid	Ansa cervicalis (C1–C3) → depresses hyoid
Sternothyroid	Ansa cervicalis (C1–C3) → depresses larynx
Styloglossus	CN XII → retracts + elevates tongue
Stylohyoid	CN VII → retracts/elevates hyoid
Stylopharyngeus	CN IX → ONLY pharyngeal muscle via CN IX; elevates pharynx
Superior oblique (eye)	CN IV (trochlear) → depresses + abducts + intorts; "down and out"
Superior rectus (eye)	CN III → elevates + adducts + intorts eyeball
Temporalis	CN V3 → elevates + retracts mandible; fan-shaped; temporal fossa
Tensor veli palatini	CN V3 → opens auditory tube; tenses soft palate
Thyrohyoid	C1 via CN XII → depresses hyoid / elevates larynx
Trapezius	CN XI + C3–C4 → elevate/retract/depress/rotate scapula
Zygomaticus major	CN VII → smiling (pulls corner of mouth up + back)
Carina (trachea)	Bifurcation T4–T5; most sensitive cough reflex area
Coronal suture	Frontal + parietals
Corpus callosum	Largest commissure; connects L ↔ R hemispheres (commissural fibers)
Cribriform plate	Ethmoid; CN I (olfactory) passes through
Crista galli	Ethmoid; falx cerebri attaches here
Ethmoid	Cribriform plate + crista galli + sup/mid nasal conchae + perp. plate (septum)
Foramen magnum	Largest foramen; inferior occipital; spinal cord + vertebral aa. pass through
Foramen ovale	Sphenoid; CN V3 (mandibular)
Foramen rotundum	Sphenoid; CN V2 (maxillary)
Foramen spinosum	Sphenoid; middle meningeal artery
Frontal bone	Forehead; frontal sinuses; supraorbital foramen (CN V1)
Hard palate	Anterior 2/3 = maxillae; posterior 1/3 = palatines
Hyoid	Only bone NOT articulating with another bone
Inferior nasal conchae	Separate bone (NOT part of ethmoid); turbulence for air warming
Internal acoustic meatus	Temporal bone; CN VII + CN VIII pass through
Jugular foramen	Occipital/temporal; CN IX, X, XI + internal jugular vein

Lacrimal bone	Smallest facial bone; medial orbit wall; nasolacrimal duct
Lambdoid suture	Parietals + occipital
Mandible	Only freely movable skull bone; mental foramen (ant.) + mandibular foramen (post. ramus)
Mastoid process	Temporal bone; attachment for SCM
Maxillae	Upper jaw; largest paranasal sinuses; infraorbital foramen (V2); ant. 2/3 hard palate
Nasal septum	Superior = perp. plate of ethmoid; inferior = vomer; anterior = septal cartilage
Occipital bone	Posterior/inferior skull; foramen magnum; occipital condyles → atlas (C1)
Optic canal	Sphenoid lesser wing; CN II + ophthalmic artery
Pituitary fossa	Sella turcica of sphenoid
Sagittal suture	R parietal + L parietal
Sella turcica	Sphenoid; houses pituitary gland
Sphenoid	Keystone of cranium; sella turcica; foramen ovale (V3); foramen rotundum (V2); optic canal
Squamous suture	Parietal + temporal
Styloid process	Temporal bone; attachment for stylohyoid, styloglossus, stylopharyngeus
Stylomastoid foramen	Temporal; CN VII exits here
Superior orbital fissure	Sphenoid; CN III, IV, V1 (ophthalmic), VI
Temporal bone	Mastoid process; styloid; ext/int acoustic meatus; houses middle + inner ear; TMJ
Vomer	Inferior nasal septum; midsagittal bone
Zygomatic bone	Cheekbone; articulates with temporal, frontal, maxillary, sphenoid (all 4)
Amygdala	Fear conditioning + emotional responses; anterior temporal lobe
Basal ganglia	Caudate + putamen + globus pallidus; modulates motor; substantia nigra degeneration → Parkinson's
Broca's area	L frontal lobe; speech PRODUCTION; damage → expressive aphasia
Calcarine sulcus	Primary visual cortex; occipital lobe
Central sulcus	Separates frontal (motor) from parietal (sensory)
Cerebellum	Coordination, balance, fine motor; spinocerebellar tracts
Corpus callosum	Connects L ↔ R hemispheres; largest commissure
CSF pathway	Lateral ventricles → 3rd ventricle → cerebral aqueduct → 4th ventricle → subarachnoid
Decussation of pyramids	Medulla; lateral corticospinal tract crosses here
Hippocampus	Memory formation/consolidation; medial temporal lobe
Hypothalamus	Controls ANS, temperature, hunger/thirst, circadian rhythm; regulates pituitary
Internal capsule	Projection fibers; cortex ↔ brainstem/spinal cord; between thalamus + basal ganglia
Pineal gland	Epithalamus; melatonin; circadian rhythm
Postcentral gyrus	Primary SENSORY cortex; parietal lobe (posterior to central sulcus)
Precentral gyrus	Primary MOTOR cortex; frontal lobe (anterior to central sulcus)
Substantia nigra	Basal ganglia; dopamine; degeneration → Parkinson's disease
Thalamus	Relay center; all sensory (except smell) passes through; diencephalon
Wernicke's area	L temporal lobe; speech COMPREHENSION; damage → receptive aphasia
CN I – Olfactory	Sensory; smell; cribriform foramina of ethmoid
CN II – Optic	Sensory; vision; optic canal (sphenoid)
CN III – Oculomotor	Motor; most eye muscles (SR, IR, MR, IO, levator); pupil constriction; superior orbital fissure
CN IV – Trochlear	Motor; superior oblique ONLY; superior orbital fissure; smallest CN; only CN from dorsal brainstem
CN V – Trigeminal	Mixed; face sensation (V1/V2/V3); mastication (V3); largest CN
CN V1 – Ophthalmic	Sensory; forehead/scalp/cornea; superior orbital fissure
CN V2 – Maxillary	Sensory; cheek/upper lip/upper teeth; foramen rotundum
CN V3 – Mandibular	Mixed; lower jaw/teeth; mastication muscles; foramen ovale
CN VI – Abducens	Motor; lateral rectus ONLY; superior orbital fissure
CN VII – Facial	Mixed; facial expression; taste ant 2/3 tongue; lacrimal/salivary glands; stylomastoid foramen
CN VIII – Vestibulocochlear	Sensory; hearing (cochlear) + balance (vestibular); internal

	acoustic meatus
CN IX – Glossopharyngeal	Mixed; taste post 1/3 tongue; parotid gland; carotid body/sinus; jugular foramen
CN X – Vagus	Mixed; 75% of parasympathetic; thorax + abdomen; larynx/pharynx; jugular foramen
CN XI – Accessory	Motor; trapezius + SCM; jugular foramen
CN XII – Hypoglossal	Motor; tongue muscles (except palatoglossus); hypoglossal canal
Ansa cervicalis	C1–C3; innervates infrahyoid muscles (sternohyoid, sternothyroid, omohyoid)
Muscarinic receptors	Parasympathetic neuroeffector junctions; blocked by atropine
Nicotinic receptors	ALL ganglionic neurons of BOTH divisions; also NMJ
Parasympathetic preganglionic	Long; craniosacral (CN III,VII,IX,X + S2–S4); ACh → nicotinic
Sympathetic postganglionic	NE (except sweat glands = ACh); alpha + beta adrenergic receptors
Cones	Color vision; concentrated in fovea centralis; 3 types (L/M/S wavelength)
Eustachian tube	Connects middle ear to nasopharynx; equalize pressure
Fovea centralis	Center of macula; highest visual acuity; cones only
Malleus, incus, stapes	Ossicles in tympanic cavity of temporal bone; amplify sound
Optic disc	Blind spot; where CN II exits; no photoreceptors
Organ of Corti	In cochlea; converts sound waves to nerve impulses (hair cells)
Rods	Low-light/peripheral vision; NOT in fovea; rhodopsin pigment
Semicircular canals	Dynamic equilibrium; rotational movement detection; crista ampullaris
Utricle + saccule	Static equilibrium; linear acceleration; macula with otoliths
Arytenoid cartilage	Hyaline; moves with cricoarytenoid joints to open/close glottis
Carina	Internal ridge at tracheal bifurcation (T4–T5); most sensitive for cough
Epiglottis	ELASTIC cartilage; covers glottis during swallowing
Glottis	Opening between true vocal folds
Laryngopharynx	From hyoid → esophagus/larynx; CN X + IX
Nasopharynx	Above soft palate; CN X + IX; eustachian tube opening; adenoids
Oropharynx	Soft palate → hyoid; CN X + IX; tonsils; food + air pathway
Right primary bronchus	More vertical + wider → foreign objects most likely lodge here
Thyroid cartilage	Largest laryngeal cartilage; HYALINE; Adam's apple
Trachea	C-shaped hyaline cartilage rings; trachealis = smooth muscle on posterior
True vocal folds	Ligamentous; vibrate for voice; closed by all laryngeal mm. except post. cricoarytenoid
Adrenal (suprarenal) cortex	Cortisol, aldosterone, androgens; three zones (G/F/R)
Adrenal (suprarenal) medulla	Epinephrine + norepinephrine; chromaffin cells; modified sympathetic ganglion
Anterior pituitary (adenohypophysis)	GH, TSH, ACTH, FSH, LH, prolactin; regulated by hypothalamic releasing hormones
Parathyroid glands	4 glands on posterior thyroid; PTH → ↑ blood Ca ²⁺ ; smallest endocrine glands
Pineal gland	Epithalamus; melatonin; circadian rhythm regulation
Posterior pituitary (neurohypophysis)	ADH (vasopressin) + oxytocin; stored/released here but produced in hypothalamus
Thyroid gland	T3/T4 (metabolism) + calcitonin (↓ blood Ca ²⁺); largest endocrine gland in neck
Basilar artery	Formed by L + R vertebral arteries joining at pons
Circle of Willis	Anastomosis: internal carotids + basilar artery; provides collateral circulation
Dural venous sinuses	Between dura mater layers; drain cerebral blood → internal jugular vein
External carotid artery	Branches supply face, scalp, neck, tongue; does NOT enter skull
Internal carotid artery	Enters carotid canal → cavernous sinus; supplies anterior brain + eye
Sigmoid sinus	Continues from transverse sinus → drains into internal jugular vein
Superior sagittal sinus	Runs in falx cerebri (superior edge); drains into transverse sinuses
Vertebral arteries	From subclavian → foramen transversarium (C6–C1) → foramen magnum → basilar

1. MUSCLES — Organized by Innervating Nerve

Find the nerve column first to answer "what innervates X?" — or scan the muscle column to confirm a nerve.

MUSCLE	NERVE (innervation)	ACTION	NOTES
Masseter	CN V3 (mandibular)	Elevates mandible (closes jaw)	Most powerful jaw closer; palpable
Temporalis	CN V3	Elevates + retracts mandible	Fan-shaped; fills temporal fossa
Medial pterygoid	CN V3	Elevates + medially deviates mandible	Deep to masseter
Lateral pterygoid	CN V3	OPENS jaw; protrudes mandible	ONLY muscle that opens jaw
Mylohyoid	CN V3	Elevates floor of mouth; depresses mandible	
Ant. digastric	CN V3	Depresses mandible; elevates hyoid	Post. belly = CN VII; 2 bellies
Tensor veli palatini	CN V3	Opens auditory tube; tenses soft palate	
Orbicularis oris	CN VII (facial)	Closes + purses lips	
Orbicularis oculi	CN VII	Closes eyelid; blinks; squints	
Buccinator	CN VII	Compresses cheeks; blowing/sucking	NOT a mastication muscle
Zygomaticus major	CN VII	Smiling (pulls mouth corner up + back)	
Platysma	CN VII	Depresses mandible; wrinkles neck skin	
Occipitofrontalis	CN VII	Raises eyebrows; wrinkles forehead	
Post. digastric	CN VII	Retracts/elevates hyoid; depresses mandible	
Stylohyoid	CN VII	Retracts + elevates hyoid	
Stapedius	CN VII	Dampens ossicle vibration (loud sounds)	In middle ear
Stylopharyngeus	CN IX (glosso.)	Elevates pharynx during swallowing	ONLY pharyngeal m. via CN IX
Pharyngeal constrictors (sup/mid/inf)	CN X (vagus)	Constrict pharynx; propel food bolus	
Palatoglossus	CN X	Elevates post. tongue; depresses soft palate	EXCEPTION: tongue m. via CN X not XII
Cricothyroid	CN X (superior laryngeal)	Tenses/lengthens vocal folds (↑ pitch)	Only laryngeal m. via SLN (not RLN)
Post. cricoarytenoid	CN X (recurrent laryngeal)	Abducts vocal folds — ONLY abductor	Damage → hoarseness + airway risk
All other laryngeal mm.	CN X (recurrent laryngeal)	Adduct/relax vocal folds	
Trapezius	CN XI (accessory) + C3–C4	Elevate/retract/depress/rotate scapula	
SCM	CN XI + C2–C3	Flex/rotate head contralateral; bilateral = flex neck	Clinical landmark
Genioglossus	CN XII (hypoglossal)	Protrudes tongue; depresses center	Largest extrinsic tongue m.
Hyoglossus	CN XII	Depresses + retracts tongue	
Styloglossus	CN XII	Retracts + elevates tongue	
Geniohyoid	C1 via CN XII	Elevates hyoid; depresses mandible	
Thyrohyoid	C1 via CN XII	Depresses hyoid or elevates larynx	
Sternohyoid	Ansa cervicalis (C1–C3)	Depresses hyoid post-swallowing	
Sternothyroid	Ansa cervicalis (C1–C3)	Depresses larynx	
Omohyoid (sup+inf)	Ansa cervicalis (C1–C3)	Depresses hyoid	2 bellies; intermediate tendon
Scalenus ant/med/post	C3–C8 (cervical plexus + brachial)	Lateral flex neck; elevate 1st/2nd rib (inspiration)	Brachial plexus passes between ant + med scalene
Superior rectus	CN III (oculomotor)	Elevates + adducts + intorts eyeball	
Inferior rectus	CN III	Depresses + adducts + extorts	

		eyeball	
Medial rectus	CN III	Adducts eyeball (medial gaze)	
Inferior oblique	CN III	Elevates + abducts + extorts (up-out)	
Levator palpebrae	CN III (+ sympathetic)	Raises upper eyelid	Damage → ptosis; Horner's = partial ptosis
Superior oblique	CN IV (trochlear)	Depresses + abducts + intorts; "down and out"	Passes through trochlea; smallest CN; dorsal brainstem
Lateral rectus	CN VI (abducens)	Abducts eyeball (lateral gaze)	Damage → esotropia (medial drift)

Memory rule: "LR6SO4" — Lateral Rectus = CN VI; Superior Oblique = CN IV; ALL OTHERS = CN III.

All mastication muscles = CN V3. All facial expression muscles = CN VII. All tongue muscles = CN XII EXCEPT Palatoglossus (CN X).

2. CRANIAL NERVES — CN I–XII Master Table

CN	NAME	TYPE	FUNCTION	FORAMEN	KEY FACT
I	Olfactory	S	Smell	Cribriform foramina (ethmoid)	Only CN that bypasses thalamus
II	Optic	S	Vision	Optic canal (sphenoid)	NOT a true CN; extension of brain
III	Oculomotor	M	SR, IR, MR, IO, levator; pupil constriction (PS)	Superior orbital fissure	Damage → "down and out" + ptosis + fixed dilated pupil
IV	Trochlear	M	Superior oblique	Superior orbital fissure	Smallest CN; ONLY CN from dorsal brainstem
V	Trigeminal	M+S	Face sensation (V1/V2/V3); mastication (V3 only)	SOF(V1), Rotund(V2), Ovale(V3)	LARGEST CN; 3 branches
VI	Abducens	M	Lateral rectus only	Superior orbital fissure	Longest intracranial course → common injury
VII	Facial	M+S+PS	Facial expression; taste ant 2/3; lacrimal, submandibular, sublingual glands	Internal acoustic meatus → stylomastoid foramen	Damage → Bell's palsy (unilateral face drop)
VIII	Vestibulocochlear	S	Hearing (cochlear br.) + Balance (vestibular br.)	Internal acoustic meatus	Cochlear → spiral organ; Vestibular → semicircular canals
IX	Glossopharyngeal	M+S+PS	Taste post 1/3 tongue; parotid gland; carotid body/sinus; stylopharyngeus	Jugular foramen	Carotid body (O2 sensor) + carotid sinus (BP sensor)
X	Vagus	M+S+PS	75% of parasympathetic; thorax + abdomen viscera; larynx/pharynx; palate	Jugular foramen	Longest CN; CN of "rest and digest"; RLN + SLN
XI	Accessory	M	Trapezius + SCM	Jugular foramen	Spinal root from C1–C5 → enters skull via foramen magnum
XII	Hypoglossal	M	Tongue muscles (all except palatoglossus)	Hypoglossal canal (occipital)	Damage → tongue deviates TO damaged side

Type key: S=Sensory, M=Motor, M+S=Mixed, PS=Parasympathetic. Mnemonics: "Some Say Marry Money But My Brother Says Big Brains Matter More" (types: S,S,M,M,M+S,M,M+S,S,M+S,M+S,M,M).

3. SKULL BONES — Key Features & Foramina

3.1 Cranial Bones (8)

BONE	KEY FEATURES	FORAMINA / STRUCTURES	EXAM TRAPS
Frontal	Forehead; frontal sinuses	Supraorbital foramen (CN V1)	Forms anterior cranial fossa floor
Parietal (×2)	Roof + sides of cranium	—	Sagittal suture between them; coronal (w/ frontal); lambdoid (w/ occipital)
Temporal (×2)	Mastoid process; styloid process; TMJ (mandibular fossa); houses middle + inner ear	Ext./int. acoustic meatus; carotid canal; jugular foramen; stylomastoid foramen (CN VII)	Petrous portion houses cochlea/vestibular apparatus
Occipital	Posterior/inferior; occipital condyles → C1 atlas	Foramen magnum (largest; spinal cord + vertebral aa.); hypoglossal canal (CN XII)	
Sphenoid	Keystone of cranium; sella turcica (pituitary); greater + lesser wings	Foramen ovale (V3); foramen rotundum (V2); optic canal (CN II + ophthalmic a.); SOF (CN III, IV, V1, VI); foramen spinosum (middle meningeal a.)	Most of middle cranial fossa floor
Ethmoid	Cribiform plate; crista galli; sup + mid nasal conchae; perpendicular plate (nasal septum)	Cribiform foramina (CN I olfactory)	Most of medial orbit wall; perp. plate = superior nasal septum

3.2 Facial Bones (14) + Fontanels

BONE	KEY FEATURES	NOTE
Nasal (×2)	Bridge of nose	
Lacrima (×2)	Smallest facial bone; medial orbit wall; nasolacrimal duct	Does NOT form cheekbone
Zygomatic (×2)	Cheekbone; orbit lateral wall; articulates with ALL 4: temporal, frontal, maxillary, sphenoid	Zygomaticofacial foramen
Maxillae (×2)	Upper jaw; largest paranasal sinuses; infraorbital foramen (V2); ant. 2/3 hard palate	Bears upper teeth
Palatines (×2)	Post. 1/3 hard palate; inferior orbit floor	
Inf. nasal conchae (×2)	Lateral nasal walls; turbulence	Separate bone — NOT part of ethmoid
Vomer	Inferior nasal septum; midsagittal	
Mandible	Only movable skull bone; mental foramen (CN V3 ant.); mandibular foramen (CN V3 int. ramus)	TMJ = synovial joint with temporal bone
Hyoid	Only bone NOT articulating with another	U-shaped; supports tongue; larynx attachment

Anterior fontanel (bregma) = largest; last to close (~18 mo). Posterior fontanel = closes first (~2 mo).

4. BRAIN REGIONS & LOBES

REGION	LOCATION/DIVISIONS	KEY FUNCTIONS	EXAM FLAGS
Frontal lobe	Anterior; anterior to central sulcus	Primary motor (precentral gyrus); Broca's (L) = speech production; prefrontal = reasoning/planning; premotor = complex motor patterns	Damage to Broca's → expressive aphasia
Parietal lobe	Posterior to central sulcus; above lateral sulcus	Primary sensory (postcentral gyrus); somatosensory association; spatial orientation; Wernicke's (L posterior)	Damage to Wernicke's → receptive aphasia
Temporal lobe	Inferior; lateral sulcus = superior border	Primary auditory (Heschl's gyrus); Wernicke's (L); memory; olfactory (uncus)	Hippocampus medially
Occipital lobe	Posterior pole; parieto-occipital	Primary visual cortex	

	sulcus (anterior)	(calcarine sulcus); visual processing	
Insula	Deep to lateral sulcus	Visceral sensation; gustatory cortex (TASTE); pain processing	Hidden by temporal/frontal/parietal opercula
Thalamus	Diencephalon; "egg-shaped" flanks 3rd ventricle	RELAY for all sensory except smell; motor relay; consciousness	Smell bypasses thalamus directly to cortex
Hypothalamus	Diencephalon; floor of 3rd ventricle	ANS control; temperature; hunger/thirst; circadian; regulates pituitary	Controls anterior pituitary via portal system; produces ADH + oxytocin
Epithalamus	Posterior diencephalon	Pineal gland (melatonin); habenula	
Pons	Brainstem	Respiration (apneustic/pneumotoxic centers); CN V,VI,VII,VIII nuclei; connects cerebellum	
Medulla oblongata	Brainstem; most inferior	Vital centers: cardiac, respiratory, vasomotor; CN IX,X,XI,XII nuclei; decussation of pyramids	Damage can be fatal
Midbrain (mesencephalon)	Brainstem	Visual (superior colliculus) + auditory (inferior colliculus) reflexes; CN III,IV; substantia nigra; red nucleus	Cerebral peduncles carry corticospinal fibers
Cerebellum	Posterior fossa; connected by 3 pairs of peduncles	Coordination; balance; fine-tune motor; DOES NOT initiate movement	Damage → ataxia, dysmetria
Basal ganglia	Cerebral hemispheres (deep white matter)	Modulates/smooths voluntary movement; habit formation; caudate + putamen (striatum) + globus pallidus + subthalamic + substantia nigra	Parkinson's = substantia nigra dopamine loss; Huntington's = striatum degeneration
Limbic system	Medial temporal + cingulate + hypothalamus	Emotions; memory (hippocampus); fear (amygdala); olfaction (uncus)	Bilateral hippocampal damage → cannot form new long-term memories

5. MENINGES, CSF & HEAD VASCULATURE

5.1 Meninges

LAYER	DESCRIPTION	KEY SPACES
Dura mater	Outermost; 2 layers (periosteal + meningeal); forms dural sinuses + folds	Epidural space (potential; blood = epidural hematoma). Folds: falx cerebri (longitudinal fissure), tentorium cerebelli (between cerebrum + cerebellum), falx cerebelli
Arachnoid mater	Middle; thin; arachnoid villi absorb CSF into dural sinuses	Subarachnoid space: filled with CSF; arachnoid trabeculae; bridging veins cross here (rupture → subdural hematoma)
Pia mater	Innermost; adheres directly to brain/cord surface; blood vessels	No space between pia and brain; denticulate ligaments (spinal cord)

5.2 CSF Pathway

Choroid plexus (lateral ventricles) → interventricular foramen of Monro → 3rd ventricle → cerebral aqueduct of Sylvius → 4th ventricle → foramina of Luschka (lateral) + Magendie (medial) → subarachnoid space → arachnoid villi → dural sinuses → blood.

5.3 Head Vasculature

VESSEL	PATH / ORIGIN	KEY FACTS
Common carotid aa.	R: brachiocephalic trunk; L: aortic arch → bifurcates at C4 (carotid sinus)	Carotid sinus (baroreceptor CN IX) + carotid body (chemoreceptor CN IX)
External carotid a.	Supplies face, scalp, neck, tongue, dura (meningeal branches)	Does NOT enter skull; 8 branches (STA = superior temporal palpable at temple)
Internal carotid a.	Enters carotid canal → cavernous sinus → anterior + middle cerebral aa.	Supplies anterior brain + eye (ophthalmic a.)
Vertebral aa.	Subclavian → foramen transversaria C6–	R + L join at pons to form basilar a.

	C1 → foramen magnum → basilar a.	
Basilar a.	Runs on pons → posterior cerebral aa.	Supplies posterior brain
Circle of Willis	ICA + basilar (+ communicating aa.) form anastomotic ring at brain base	Collateral circulation if one vessel blocked; anterior + posterior communicating aa.
Superior sagittal sinus	In superior edge of falx cerebri	Drains to transverse sinuses → sigmoid sinus → internal jugular vein
Cavernous sinus	Lateral to sella turcica	CN III, IV, V1, V2, VI pass through; ICA runs through it; infection risk (dangerous area of face)
Dural sinuses (general)	Between dura mater layers	Drain cerebral veins; ultimately → internal jugular vein

6. SPECIAL SENSES

6.1 Ear

STRUCTURE	LOCATION	FUNCTION
Auricle (pinna)	External ear	Collects + funnels sound
Tympanic membrane	Between ext. + middle ear	Vibrates with sound waves; transmits to ossicles
Ossicles: Malleus, Incus, Stapes	Middle ear (tympanic cavity in temporal bone)	Amplify + transmit vibrations: TM → oval window; M → I → S
Oval window	Medial middle ear wall	Stapes footplate → oval window → perilymph vibrates
Eustachian (auditory) tube	Middle ear ↔ nasopharynx	Equalizes middle ear pressure; connects to nasopharynx
Cochlea	Inner ear	Hearing; contains organ of Corti; basilar membrane vibrates → hair cells → CN VIII cochlear
Organ of Corti	Scala media of cochlea	Mechanoreceptors (hair cells) convert sound → nerve impulse → CN VIII
Semicircular canals (3)	Inner ear; 3 planes	DYNAMIC equilibrium (rotational); crista ampullaris + cupula detect angular acceleration
Utricle	Inner ear; vestibule	STATIC equilibrium; linear acceleration + gravity; macula + otoliths
Sacculle	Inner ear; vestibule	STATIC equilibrium; vertical acceleration

6.2 Eye

STRUCTURE	LOCATION	FUNCTION
Cornea	Anterior transparent coat	Refracts light (most refraction); avascular; CN V1 sensation
Iris	Colored; in anterior chamber	Controls pupil size; constriction = PS (CN III); dilation = sympathetic
Lens	Behind iris; biconvex	Accommodation (near vs. far focus); held by ciliary zonules
Ciliary body	Between iris + choroid	Produces aqueous humor; ciliary muscle (CN III PS) → accommodation
Aqueous humor	Anterior + posterior chambers	Nourishes cornea/lens; drains via canal of Schlemm; blockage → glaucoma
Vitreous humor	Posterior segment	Maintains eye shape; gel-like; NOT reproduced
Retina	Innermost layer; neural	Contains rods + cones; fovea centralis (highest acuity; cones only)
Rods	Peripheral retina	Scotopic (dim light) vision; rhodopsin; no color
Cones	Fovea centralis	Photopic (color) vision; 3 types (red/green/blue wavelength)
Optic disc	Where CN II exits	Blind spot; NO photoreceptors here
Fovea centralis	Center of macula lutea	Highest visual acuity; cones only
Optic chiasm	Below hypothalamus	Nasal fibers cross to contralateral optic tract; temporal fibers stay ipsilateral

6.3 Taste & Smell

SENSE	RECEPTOR	NERVE	PATH
Taste (gustation)	Taste buds on tongue papillae (fungiform, circumvallate, foliate)	Ant. 2/3 tongue = CN VII (chorda tympani); Post. 1/3 = CN IX; Epiglottis/laryngopharynx = CN X	→ Nucleus solitarius (medulla) → thalamus → insula (gustatory cortex)
Smell (olfaction)	Olfactory epithelium (superior nasal cavity)	CN I → cribriform foramina → olfactory bulb → olfactory cortex	Only sense that bypasses thalamus; directly to limbic cortex

7. ANS — Head & Neck Parasympathetic Outflow

CN	GANGLION	TARGET ORGAN	EFFECT
CN III (oculomotor)	Ciliary ganglion	Pupil (sphincter); ciliary muscle	Pupil constriction; accommodation
CN VII (facial)	Pterygopalatine ganglion (lacrimation + nasal)	Lacrimal gland; nasal mucosa; palate	Tear production; nasal secretion
CN VII (facial)	Submandibular ganglion	Submandibular + sublingual glands	Saliva production
CN IX (glossopharyngeal)	Otic ganglion	Parotid gland	Saliva (largest salivary gland)
CN X (vagus)	Terminal/intramural ganglia (throughout thorax + abdomen)	Heart (↓HR); lungs (bronchoconstriction); GI (↑motility)	75% of total parasympathetic output

8. RESPIRATORY TRACT (Upper) — Pharynx, Larynx, Trachea

REGION/STRUCTURE	EXTENT	PATHWAY	KEY FACTS
Nasopharynx	Post. to nasal cavity; above soft palate	Air only	Eustachian tube opens here; adenoids (pharyngeal tonsils)
Oropharynx	Soft palate → hyoid bone	Food + air	Palatine tonsils; CN IX (sensation); CN X (motor)
Laryngopharynx	Hyoid → esophagus/larynx inlet	Food + air	CN IX + X; narrowest part → most common food lodging site
Epiglottis	Laryngeal inlet cover	Covers glottis during swallowing	Elastic cartilage (only laryngeal elastic cartilage)
Thyroid cartilage	Largest laryngeal cartilage	Anterior protection	Hyaline; Adam's apple; no posterior wall
Cricoid cartilage	Only complete ring	Below thyroid cartilage	Hyaline; complete ring; only complete cartilage
Arytenoid cartilages	Posterior larynx (×2)	Rotate to open/close glottis	Hyaline; articulate with cricoid; pivot for vocal fold abduction/adduction
Corniculate + Cuneiform	Small cartilages atop arytenoids	Support aryepiglottic folds	Elastic cartilage
True vocal folds (ligamentous)	Glottis	Vibrate to produce voice	Closed by all laryngeal mm. except post. cricoarytenoid (abducts)
Vestibular folds (false)	Above true folds	No phonation	Protect; do not vibrate
Trachea	C6 → carina (T4–T5)	Air only	16–20 C-shaped hyaline rings; trachealis (smooth m.) on posterior; foreign objects → R bronchus

9. ENDOCRINE SYSTEM

GLAND	HORMONES	TARGET / EFFECT	KEY FACT
Anterior pituitary (adenohypophysis)	GH, TSH, ACTH, FSH, LH, Prolactin	Multiple targets; controlled by hypothalamic releasing/inhibiting hormones	Regulated by portal system from hypothalamus; no direct neural connection
Posterior pituitary (neurohypophysis)	ADH (vasopressin) + Oxytocin	ADH → kidney water reabsorption (↑ plasma osmolarity); Oxytocin → uterine contraction + milk ejection	Hormones PRODUCED in hypothalamus; STORED/RELEASED from posterior pituitary

Thyroid (bilobed; largest neck endocrine)	T3 + T4 (thyroxine) → ↑ metabolism + heat; Calcitonin → ↓ blood Ca ²⁺	Metabolic rate; growth; development	Follicular cells = T3/T4; parafollicular (C) cells = calcitonin
Parathyroid (4 glands; posterior thyroid)	PTH (parathyroid hormone) → ↑ blood Ca ²⁺	Bone resorption ↑; kidney Ca ²⁺ reabsorption ↑; activates vitamin D	Smallest endocrine glands; ANTAGONIST to calcitonin
Suprarenal cortex (outer)	Cortisol (zona fasciculata); Aldosterone (zona glomerulosa); Androgens (zona reticularis)	Cortisol → stress response, anti-inflammatory; Aldosterone → Na ⁺ retention → ↑ BP	Zones: "GFR" = Glomerulosa, Fasciculata, Reticularis
Suprarenal medulla (inner)	Epinephrine (80%) + Norepinephrine (20%)	Fight-or-flight amplification; ↑HR, ↑BP, bronchodilation	Modified sympathetic ganglion; chromaffin cells; hormones remain elevated 10–30× longer than neural effect
Pineal gland (epithalamus)	Melatonin	Circadian rhythm; sleep-wake cycle	Secretion ↑ in darkness; ↓ in light
Pancreas (islets of Langerhans)	Insulin (beta cells) → ↓ glucose; Glucagon (alpha cells) → ↑ glucose; Somatostatin → inhibits both	Blood glucose regulation	Endocrine = islets; Exocrine = acinar cells (digestive enzymes)

TB HOT SPOTS — HEAD & NECK

Facts from the Stuvia TB that need quick reference — not fully covered above.

HEAD & NECK	
TERM / CONCEPT	KEY FACT (answer the TB is looking for)
Telencephalon	Becomes the cerebrum
Metencephalon	Becomes cerebellum + pons
Diencephalon	Thalamus + hypothalamus + pineal + pituitary stalk
Mesencephalon	Midbrain: superior/inferior colliculi, cerebral peduncles, CN III/IV
Myelencephalon	Becomes medulla oblongata
BBB primary purpose	Isolate CNS from general circulation; endothelial tight junctions + astrocytes
Sagittal + coronal early closure	Causes abnormal skull growth problems
Frontal (metopic) suture	Infant skull; may be absent/unfused; located between two frontal halves
Mental protuberance	Bony prominence forming the chin; prominent = "large chin" appearance
Hyoid bone	Only bone not articulating with another; can fracture in strangulation
Sphenoid: condylar processes	FALSE — condylar processes = occipital bone, NOT sphenoid
Sacrum	5 fused sacral vertebrae
Lordosis	Exaggerated lumbar curve; caused by pregnancy or obesity
Diaphragm	Separates thoracic + abdominopelvic cavities; innervated by phrenic nerve (C3-C5)
Phrenic nerve	C3-C5; innervates diaphragm; "C3,4,5 keep the diaphragm alive"
Perineum	Two triangles: urogenital triangle (anterior) + anal triangle (posterior)

Mentalis	Muscle at tip of chin; wrinkles chin skin
Ischiocavernosus	Most anterior + lateral of urogenital triangle; compresses base of penis/clitoris
Transversus abdominis	DEEPEST abdominal muscle layer; compresses abdomen
Spinales	Erector spinae group NEAREST vertebral column (medial column)
Iliocostalis	Most POSTERIOR erector spinae column; lateral
Epicranius/Occipitofrontalis	Two bellies (frontal + occipital) connected by galea aponeurotica (collagenous sheet)
Axial muscle groups	Classified by LOCATION + FUNCTION
Internal intercostal	Aids FORCED EXPIRATION (depresses ribs); interchondral aids inspiration
Rectus abdominis	Flexes vertebral column; vertical muscle with tendinous inscriptions
Head nodding (falling asleep)	Capitis + cervicis muscle groups primarily involved
Cuspids	Also called canines; 4 total (2 upper, 2 lower)
Goblet cells in bronchi	Disappear as bronchial tree branches distally toward bronchioles
Laryngopharynx lining	Stratified squamous epithelium (like esophagus)
Brain lateralization (math)	Analytical region in LEFT hemisphere in ~90% of people regardless of handedness
Cerebrum: complex tasks	Scanning + electrical studies show multiple areas needed for complex sensory/motor
Appendicular muscles	Shoulder, pelvis, upper + lower limbs